Maincrafts Technology

AI & Machine Learning Internship – Task 2

Feature Engineering, Model Optimization & Performance Comparison

1. Introduction

This report documents the methodology, results, and conclusions of Task 2 of the Maincrafts Technology AI/ML internship. Building on Task 1 (basic model training), Task 2 explores a professional ML pipeline: feature scaling, multi-model training, quantitative comparison, and model selection.

2. Dataset

The **California Housing Dataset** (20,640 samples, 8 features) was used. The target variable is the *Median House Value*. Key input features include:

- MedInc – Median income of households
- HouseAge – Median age of houses in block
- AveRooms – Average rooms per household
- AveBedrms – Average bedrooms per household
- Population – Block population
- AveOccup – Average occupancy
- Latitude / Longitude – Geographic location

3. Methodology

Step 1 – Data Loading

Loaded the dataset into a pandas DataFrame and inspected shape, data types, and missing values. No missing values were found.

Step 2 – Feature Separation

Split the DataFrame into feature matrix X (8 columns) and target vector y (HousePrice).

Step 3 – Feature Scaling

Applied StandardScaler so each feature has mean = 0 and std = 1. This prevents high-magnitude features from dominating gradient-based models.

Step 4 – Train/Test Split

Used an 80/20 split (random_state=42): 16,512 training samples, 4,128 test samples. The test set simulates unseen data.

Step 5 – Model Training

Trained three regression algorithms: Linear Regression (baseline), Ridge Regression (L2 regularisation, alpha=1.0), and Decision Tree Regressor (max_depth=5 to prevent overfitting).

Step 6 – Evaluation

Computed RMSE (Root Mean Squared Error) and R^2 Score on the held-out test set for each model. Lower RMSE = better accuracy; higher R^2 = better explanatory power.

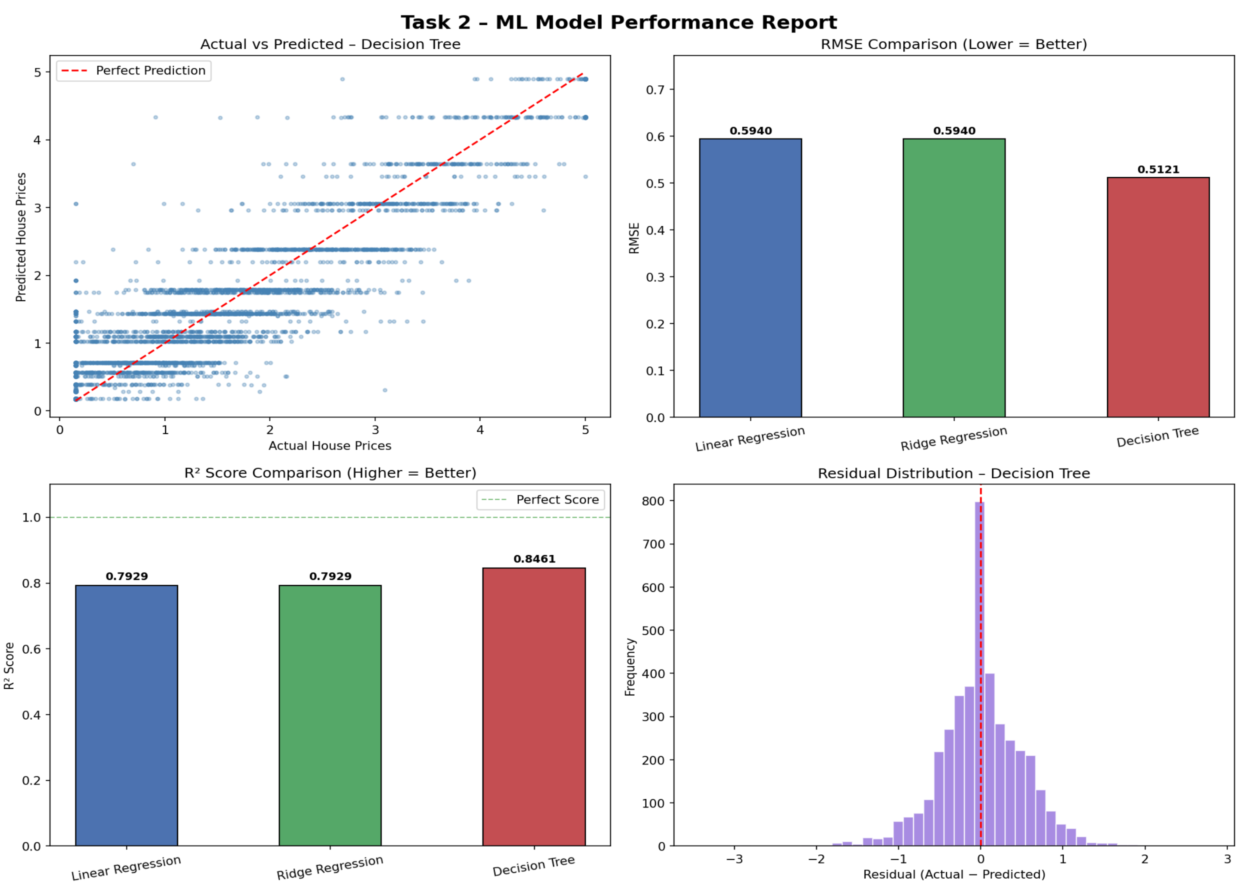
4. Results – Model Performance Comparison

Model	RMSE	R^2 Score	Notes
Linear Regression	0.5940	0.7929	Baseline – linear relationships only
Ridge Regression	0.5940	0.7929	Similar to LR; data has minimal multicollinearity
Decision Tree	**0.5121**	**0.8461**	**Best – captures non-linear patterns**

5. Visual Analysis

The chart below shows four key diagnostics generated during evaluation:

- Top-left: Actual vs Predicted scatter – points close to the red diagonal indicate accurate predictions.
- Top-right: RMSE bar chart – Decision Tree achieves the lowest error.
- Bottom-left: R^2 bar chart – Decision Tree explains 84.6% of variance.
- Bottom-right: Residual histogram – near-normal distribution centred at zero confirms model validity.



6. Conclusion & Model Selection

The **Decision Tree Regressor (max_depth=5)** is selected as the best-performing model for the California Housing prediction task.

Justification:

- Lowest RMSE (0.5121) → highest prediction accuracy on unseen data.
- Highest R^2 score (0.8461) → explains 84.6% of variance in house prices.
- max_depth=5 limits overfitting while retaining enough complexity to capture non-linear patterns (e.g. location + income interactions).
- Linear and Ridge Regression produce identical scores here, confirming minimal multicollinearity; their lower R^2 (0.79) shows linear models miss important non-linear signals.
